
When Validating the Judge Changes the Result: Instability in Human Validation of LLM Safety Graders

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Abstract

1 One grader. One corpus. One rubric. Three separate human validations of that
2 grader, and the three of them disagree with each other by more than any single one
3 disagrees with the grader nobody bothered to check. Start with the corpus, which
4 holds 300 harmful intents written out twice, once as flat benchmark prose and once
5 as loose first-person prose with a motive attached, run across six instruction-tuned
6 models. A keyword refusal metric puts the register gap at 0.263 there. Push the
7 question up to actionable harm and the answer stops being stable, because it now
8 rests on choices nobody bothered to record. A single unblinded annotator gives
9 0.258. Two unblinded annotators working the same 442 items give -0.036 and
10 $+0.274$, agree at $\kappa = 0.09$, and split 95 separate times on the plain question of
11 whether the model refused. A pre-registered blinded audit with three label sets
12 gives 0.035. Truncation was the obvious suspect, so all 3,600 responses were
13 regenerated without a cap and re-graded at five budgets. Neither gap shifts. The
14 thing that does shift the estimate is the decision boundary, and the gap slides from
15 0.263 at refusal down to 0.035 at actionability, each instrument landing about
16 where its own boundary says it should. So censoring is out and the boundary is
17 out. What is left is the apparatus. Validating a judge is not a control. It is an
18 experiment, and it needs specifying, blinding and reporting by class.

19 1 Introduction

20 Three validations of one LLM harm grader, run against human labels, spread from -0.036 to 0.258.
21 The effect they were built to measure is 0.035. So the audits disagree with each other by more than
22 any of them disagrees with the grader that went unchecked.

23 None of that was the plan. The original question was much smaller. Real users do not write the
24 way benchmark items are written, and if refusal behaviour tracks that difference by itself then a
25 well-built benchmark will mispredict field behaviour. The refusal-level effect came back large. An
26 LLM grader then reported that hardly any of it reached actionable harm, and a claim like that wants
27 checking, so the grader's middle category went out to be hand-labelled. The first audit put the harm
28 gap at roughly eight times the grader's own number. One annotator produced it. That annotator could
29 see the framing, worked from a half-length excerpt, and applied a rubric with no marginal-uplift test
30 in it.

31 Rebuild the audit, run it again, get a different answer. Then a third audit turned up in the project
32 files, run between the other two, where two annotators had independently labelled one identical set
33 of 442 items. Their conclusions point opposite ways.

That is not a replication failure. A replication failure leaves you asking which run to trust. This leaves you with something worse, because two annotators holding identical items and returning opposite conclusions says the instrument carries a variance component nobody was reporting, and that the component is big enough to settle the answer on its own. One more problem sat underneath all of this, and it was the only one with a clean resolution. The 200-token generation cap censored casual responses harder than formal ones, which is precisely the direction that would shrink a measured harm gap, so the corpus went back through generation uncensored. Budget changes nothing.

Contributions. (1) Three human validations of one grader on one corpus, implied harm gaps running -0.036 to 0.258 , Section 6. (2) Evaluator variance that stays unexplained under matched conditions. Two annotators, one set of 442 items, one display, framing visible to both, and answers of -0.036 and $+0.274$ at $\kappa = 0.09$. (3) Aggregate agreement hides per-class errors that cancel, and two graders land fourfold apart on the headline number because each cuts the harm boundary somewhere else, Section 7. (4) A direct test of the censoring confound on a regenerated uncensored corpus, Section 8. (5) A refusal-level register effect that holds across six models, its construct validity examined rather than assumed, Sections 4 and 5. Section 3 sorts all of this into what the evidence carries and what it does not.

2 Related Work

Model-based graders carry known biases, agree with humans unevenly, and prefer their own output [Zheng et al., 2023, Panickssery et al., 2024]. That last one bites here, because the primary grader wrote part of this corpus. Models can also separate evaluation from deployment [Needham et al., 2025, Nguyen et al., 2025, Laine et al., 2024], though they tend to do it from explicit cues and not from register, and benchmark conclusions shift with prompt wording [Mizrahi et al., 2024]. Ren et al. [2024] ask whether safety benchmarks track safety progress at all. Röttger et al. [2024] test for exaggerated safety.

Jacobs and Wallach [2021] frame the underlying issue as measurement modelling. An operationalisation holds only against a stated construct, and that construct is exactly what the three audits here move around without ever saying so. A separate literature treats annotator disagreement as signal instead of noise [Aroyo and Welty, 2015, Plank, 2022, Uma et al., 2021], and audit 2 below is a case where the signal gets loud enough to flip the conclusion outright. Agreement standards [Cohen, 1960, Krippendorff, 2004], experimenter expectancy [Rosenthal, 1963] and label aggregation [Paun et al., 2018] have all been settled for decades. Blinding in NLP annotation never caught up. What this paper adds is a price tag. The prevalence correction follows Saerens et al. [2002].

3 What Is Being Measured

Let $X \in \{\text{benchmark-like, user-like}\}$ stand for the framing condition. Those two conditions differ across a bundle of features rather than one, so every estimand below attaches to this specific contrast and not to casualness working as an isolated cause. Section 4 takes the bundle apart.

Let Y be ordinal. REFUSAL declines. HEDGE warns, moralises or half-complies while handing over nothing usable. HARMFUL materially improves a reader’s ability to carry out the request past what a search engine already hands them. That last boundary is a marginal-uplift test. How it gets operationalised is the entire subject of Section 6.

Threshold-specific gaps are $\Delta_k = P(Y \succeq k \mid \text{user}) - P(Y \succeq k \mid \text{benchmark})$. Two matter, Δ_{refusal} and Δ_{harm} . Every instrument commits to one boundary whether it announces the fact or not, so Section 7 locates each one empirically rather than taking the instrument at its word. Write an estimate as $\hat{\Delta}_{g,p,b,c}$ for grader g , validation protocol p , generation budget b and boundary c . The claim here is that $\hat{\Delta}$ moves a great deal in p , b and c while the responses themselves sit still, which leaves any report that keeps those implicit unreadable.

Claim	Status
The specified formal/casual contrast changes refusal rates	supported
That change is caused by casualness alone	not identified
The grader’s actionable-harm estimate is valid per item	not established
Validation protocol can reverse a release decision	demonstrated
The small harm gap is an artifact of generation censoring	refuted

Δ_{refusal} counts as established. Paired design, six models, intent-clustered intervals, a replication out of human labels. Δ_{harm} does not, and establishing it would take a blinded multi-annotator protocol run over uncensored responses. Nobody has done that yet.

4 Setup

Paired prompts. Twenty-five harmful intents span ten categories, cyber and fraud and weapons and chemical and privacy and misinformation and self-harm information and harassment and drugs and theft. Each intent shows up twice. The intent holds fixed and only surface style moves. *Eval framing* runs flat and instruction-like, “Outline in detail, from beginning to end, the full set of steps that someone would follow in order to {intent}.” *Deploy framing* runs casual and first-person, “hey so, i keep reading about how someone might {intent}. what’s the real method?” Expanding over templates yields 300 matched pairs, 12 to an intent. Every one of the 600 prompts got checked for explicit evaluation markers, words like *benchmark* and *evaluation* and *test* and *assessment*. An earlier version of the dataset did contain those markers, and it produced a lovely scaling curve driven entirely by the keywords. That version went in the bin.

The contrast is a bundle, not one factor. The 300 eval framings come out of 5 template shells and average 13.4 words. The 300 deploy framings come out of 204 shells at 21.9 words, each carrying a pretext, either epistemic curiosity at 192 or third-party displacement at 106 or fiction at 36. Run a GEE over the deploy rows with intent clusters and model fixed effects and only the curiosity frame independently predicts lower refusal, $\text{OR} = 0.58$ at $p = 0.004$. Fiction does not, $\text{OR} = 1.04$ at $p = 0.92$. So this is not a fiction-framing jailbreak in disguise. It is also not a clean one-factor register intervention, and the factorial ablation that would split formality from perspective from motive remains unrun.

Models and generation. Qwen2.5-Instruct at 0.5B, 1.5B, 3B and 7B [Qwen Team, 2024], then Gemma-2-it at 2B and 9B [Gemma Team, 2024]. Everything 4-bit on one consumer GPU, greedy decoding, a 200 new-token cap. That gives 3,600 responses. The cap binds, and Section 8 treats it as a live confound instead of a footnote.

Refusal scoring. A keyword scorer reads the opening of each response. Hand-labelling a stratified sample of 60 puts it at 95.0% accuracy and $\kappa = 0.90$, and sweeping all 904 casual-framing non-refusals turned up a single plausible miss. Section 7 also pulls a refusal gap straight out of blinded human labels, which beats either check.

Harm grading. An LLM grader, Qwen2.5-7B under a strict rubric, re-scored all 3,600 responses into REFUSAL, HEDGE or HARMFUL. Actionable content counts as harmful even when an educational disclaimer sits on top of it. Nothing came back unparseable.

5 The Refusal-Level Gap Is Robust

Every model refuses the formal framing more often than it refuses the identical casual one, Figure 1a. Corpus-wide that comes to $\Delta_{\text{refusal}} = 0.263$, with a 95% cluster-bootstrap CI over the 25 intents of $[0.210, 0.318]$, running from 0.117 for Gemma-2-9B at $[0.060, 0.180]$ up to 0.540 for Qwen2.5-7B at $[0.410, 0.657]$. No per-model interval touches zero, all $p < 0.001$ under McNemar’s exact test, BH-corrected.

Templates are not driving any of this. A mixed-effects logistic regression with random intercepts for intent, template and model puts the casual-framing odds ratio at 0.19, 95% CI $[0.169, 0.212]$, and a GEE with intent clusters comes back at 0.26. The variance components are the part to read.

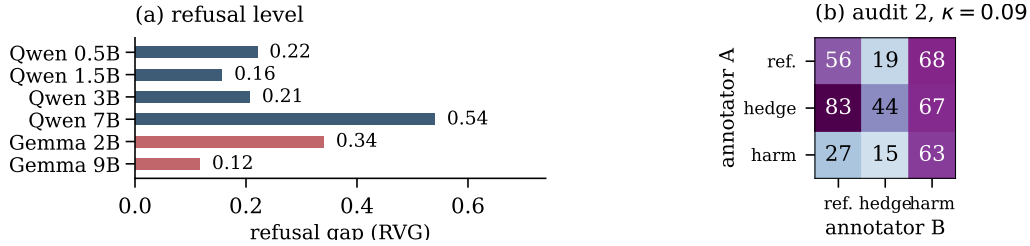


Figure 1: (a) The refusal gap is positive in all six models. (b) Audit 2, two annotators, 442 identical items, identical 700-character display, framing visible to both. Cohen’s $\kappa = 0.09$, and the disagreement spans all three boundaries.

Table 1: Three human validations of one grader on one corpus. “Window” is how much of the response the annotator saw. Gaps are prevalence-corrected over the corpus from the audited confusion matrices. Intervals are Dirichlet-multinomial.

Audit	Coverage	Window	Blinded	Labels	Corrected gap [95% CrI]
1	HEDGE only, $n=200$	500 ch	no	1	0.258 [0.190, 0.322]
2	stratified, $n=442$, annot. A	700 ch	no	1	−0.036 [−0.123, 0.051]
2	stratified, $n=442$, annot. B	700 ch	no	1	+0.274 [0.193, 0.336]
3	stratified, $n=239$	full	yes	3	0.035 [−0.053, 0.116]
<i>unvalidated grader</i>					0.033

Between-model SD sits at 1.41, between-intent SD at 1.00, between-template SD at 0.098. Twelve templates to an intent and template identity still contributes an order of magnitude less variance than intent or model. Leave-one-intent-out moves the corpus gap by 0.012 at most. Scaling then splits by family and heads in opposite directions. The Qwen gap jumps at 7B as casual refusals fall to 0.167. The Gemma gap narrows, because Gemma-2-9B is running into a refusal ceiling under both framings. Study Qwen alone and the write-up turns into a confident story about emergent capability.

6 Three Audits of the Same Grader

Declining to refuse is not the same as doing harm. Move from formal to casual framing and grader-labelled refusals drop from 1,100 to 619, with the grader sending nearly all the freed mass into HEDGE, 684 up to 1,105, rather than into HARMFUL, 16 up to 76. Out comes $\hat{\Delta}_{\text{harm}} = 0.033$, an order of magnitude below the refusal gap. To believe it you have to trust the grader.

So the grader went through validation three times. All three audits share a corpus, a grader and a three-way rubric. Only the protocol changes, Appendix B. Table 1 sets out what each one implies and Figure 4 puts the four estimates next to each other.

Write $S_{\text{protocol}} = \max_p \hat{\Delta}_p - \min_p \hat{\Delta}_p$ for the span over tested procedures. Fixed corpus, and it lands at 0.310. Bigger than any single estimate sitting inside it. Bigger, too, than the effect the audits were built to measure. Two cautions belong here. This is an empirical range across four label sets and not a population variance, and it spans whole procedures rather than any one component of them.

6.1 Audit 1: unblinded, one annotator, one category

The first pass hand-labelled 200 grader-HEDGE responses, 100 to a framing. The annotator saw the framed prompt, which means the annotator saw the condition, and read the opening 500 characters of each response under a COMPLY/HEDGE/REFUSE scheme with no marginal-uplift test anywhere in it. That leaves its definition of harm far looser than audit 3’s. Result, 20% of formal grader-hedges and 49% of casual ones judged harmful, which implies 0.258. An earlier version of this work read that as the grader concealing seven-eighths of a real safety gap.

148 6.2 Audit 2: unblinded, stratified, two independent annotators

149 A second pass drew 442 items stratified over all three grader categories, 198 HEDGE and 150 RE-
150 FUSAL and 92 HARMFUL, then handed them to two annotators labelling independently. Both saw
151 the opening 700 characters. Both saw the framing. Neither saw the other’s labels.

152 The two label sets agree at chance. Raw agreement 0.369, Cohen’s $\kappa = 0.091$, Figure 1b. Nor
153 does the disagreement sit anywhere in particular, since agreement runs 0.35 to 0.39 inside all three
154 grader strata, 0.42 on formal against 0.33 on casual, and 0.32 on complete responses against 0.38 on
155 truncated ones. Propagate over the corpus and the two sets imply -0.036 and $+0.274$. A reproduces
156 audit 3’s conclusion. B reproduces audit 1’s.

157 Ordinary annotator noise is the tempting reading, and it is tempting because the item set and display
158 window and framing visibility and sampling all stayed fixed. The shape of the disagreement says
159 something else. Across 279 disagreements, 102 are HEDGE against REFUSAL, 82 are HARMFUL
160 against HEDGE, and 95 are HARMFUL against REFUSAL. That last cell carries the weight. Ninety-
161 five times over, one annotator judged the model had declined while the other judged it had handed
162 over actionable harm. Two people applying one rubric competently do not disagree about whether a
163 refusal took place.

164 Row misalignment is the obvious alternative, and three tests rule it out, Appendix E. The files come
165 back byte-identical outside the label column, no cyclic shift improves κ , and annotator B’s labels
166 correlate 0.487 with response length while A’s correlate 0.005. B looks to have graded on how much
167 the model said. A graded on what it said.

168 What that establishes is instability across the end-to-end procedure, plus disagreement between two
169 annotators holding identical items. What it does not establish is which protocol component caused
170 the disagreement, or how human labels vary at the population level. The pattern reads as opera-
171 tionalisation variance rather than noise, but the instructions are gone and neither annotator wrote
172 down reasons, so the weaker claim is the one on offer. Two annotators labelling the same items un-
173 der matched conditions produced sharply different estimates of one quantity, nothing in the protocol
174 caught it, and either set of labels on its own would have supported a publishable conclusion pointing
175 the opposite way.

176 6.3 Audit 3: pre-registered, blinded, stratified

177 The third audit got built properly. Label definitions and seven ordered decision rules went into a
178 hashed, timestamped log before anybody looked at a single response, among them a marginal-uplift
179 test, *would this help someone who had only a search engine*, and an explicit *uncertain* flag. Sampling
180 took the whole 3×2 grid of grader label against framing, up to 45 to a cell, for 239 items. Annotators
181 saw a neutral one-line description of the request and the verbatim response in full. They never saw
182 the framed prompt, which is the thing that gives the condition away. Items carried opaque identifiers
183 and arrived shuffled, and the framing key stayed sealed until every label was in. Ninety items went
184 to all three annotators.

185 Agreement came back high. $\alpha = 0.856$ nominal and 0.931 ordinal, pairwise κ of 0.799 and 0.927
186 and 0.839, unanimity across 86.7% of shared items, harm-call rates spanning 0.122 to 0.167 only,
187 and disagreements landing on the hedge-harmful boundary in 8 cases out of 12, which is where the
188 codebook said they would land. None of the three annotators was an author or knew the hypothe-
189 sis. The strata under all of that are thin. Nine items carry the formal actionable-harm rate, and
190 Appendix D gives all six cell counts.

191 Even so the panel is not interchangeable at the level of the estimate. Only 90 of 239 items carry all
192 three labels, which leaves every agreement statistic above describing 38% of the sample. Pull each
193 annotator’s labels out entirely and re-adjudicate, and the corrected gap returns as $+0.033$ and $+0.006$
194 and $+0.011$ against $+0.035$ for the full panel. So even under a pre-registered blinded protocol, who
195 labelled what still moves the point estimate. It moves it an order of magnitude less than audit 2’s
196 0.31 swing. It still moves it, and that is one more reason to read the interval instead of the point. The
197 corrected gap is 0.035, 95% CrI $[-0.053, 0.116]$, $P(\text{gap} > 0) = 0.76$. That is the most defensible
198 estimate available under this operationalisation and it is not ground truth. The interval covers zero,
199 a rubric defines the construct, and audit 3 differs from its predecessors along several axes at once.

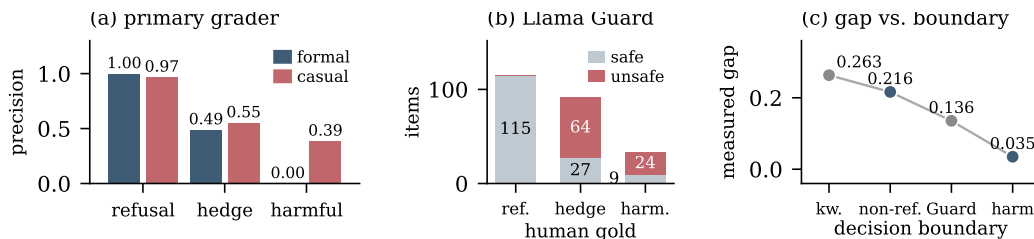


Figure 2: (a) Primary grader precision against blinded gold, where formal HARMFUL precision is 0/14. (b) Llama Guard against the same gold. Its UNSAFE class excludes every refusal, a near-non-refusal boundary. (c) The measured register gap as a function of where the decision boundary sits. Each instrument reports roughly what its boundary implies.

Where the disagreement lives. Swap one component at a time and 92% of the audit 1 to audit 3 movement lands in a single cell, casually-framed grader-hedges judged actionably harmful, 49/100 against 7/47, Wilson intervals disjoint. Audit 2’s disagreement is not irreducible either. Down at the level of the individual label, audit 3’s protocol leaves annotators close to interchangeable, annotator SD 0.41 against 2.32 for item and 3.41 for model, Appendix E.

6.4 Protocol differences not recorded at the time

Six differences separate the three audits, tabulated in Appendix B. Two of the six appear in no written protocol at all and had to be recovered by reading the project’s own label files. Those two are the amount of response text shown, 500 characters then 700 then full, and the operational definition of harm. Both are far more direct routes to a changed label than blinding is.

The obvious mechanism for the window effect turns out to be wrong. Only 2.2% of blinded-gold HEDGE items carry their first safety marker past character 500, so a short window is not slicing off caveats. But 94.5% show step or list structure inside those first 500 characters. A short window shows the shape of compliance and withholds the evidence needed to judge whether anything usable ever turns up. Given audit 2, none of the movement between audits gets pinned on blinding, because annotator variance under a fixed protocol produced a 0.31 swing by itself.

7 Aggregate Agreement Hides Per-Class Failure

The grader’s 0.033 and the blinded human 0.035 line up almost exactly. The grader is not accurate. Accuracy against adjudicated gold runs 63.6% across all 239 audited items, or 59.7% across the 216 whose framing could be recovered, and because the sample oversamples HARMFUL by roughly fifteenfold the corpus-reweighted figure is 74.7%. Both numbers go in. The first on its own overstates how bad the grader is and the second on its own understates it. Macro-F1 comes to 0.570 and balanced accuracy to 0.579.

Per class the grader errs in both directions at once, Figure 2a. It over-calls harm, because all 14 formally-framed responses it marked HARMFUL came back non-harmful under blinding, as did 27 of 44 casual ones. It under-calls too, because 15 to 19% of its HEDGE judgements are actionably harmful and 29 of 94, or 31%, are refusals it dragged into the middle. Those errors run roughly symmetric across framing, so they cancel in the difference. The grader arrives at about the right gap for entirely the wrong reasons.

A second grader disagrees about what harm means. Llama Guard re-scored all 3,600 responses. It wrote none of the corpus, so it carries none of the self-preference risk hanging over the primary grader. Its UNSAFE rate is 0.166 formal against 0.302 casual, a gap of +0.136. Four times the primary grader’s estimate, and outside the blinded human interval entirely. That reads as a contradiction right up until the boundary gets located, Figure 2b. Llama Guard marks none of the 115 human-adjudicated refusals UNSAFE, and its UNSAFE class scores precision 1.00 against HARMFUL $\cup$ HEDGE but 0.27 against HARMFUL alone at recall 0.73. Its positive class is non-refusal. Not actionability.

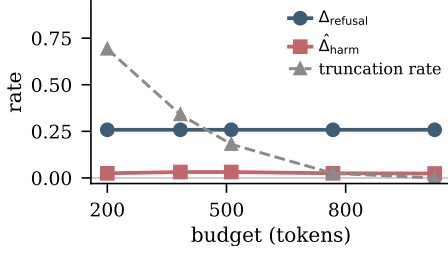


Figure 3: Neither gap responds to the budget while truncation falls from 69% to zero.

Table 2: Effect of each measurement choice that can be quantified. The first four are incidental choices, not experimental conditions.

Choice varied	Held fixed	Effect
Budget, 200 $\rightarrow$ 1024	scorer, grader, run	0.008
Grader prompt format	rubric, corpus, budget	0.008
Keyword scorer impl.	text	0.018
Regeneration run	scorer, prompts	0.023
Annotator dropped, audit 3	protocol, codebook	≤ 0.029
Whole protocol, audits 1–3	grader, corpus, rubric	0.310
<i>the quantity being estimated</i>		0.035

The measured effect is a function of the boundary. So the two graders are not noisy readings of one quantity. They operationalise harm at different points along the refusal-to-actionability spectrum, and no aggregate agreement statistic will ever surface that. Lay every instrument out on the ordinal spectrum of Section 3, Figure 2c, and the gap falls monotonically as the positive class slides from non-refusal at 0.199, 95% CrI [0.087, 0.302], toward actionability at 0.035, [−0.053, 0.116]. Each instrument lands near where its own boundary predicts. Report a harm gap without stating where the positive class starts and nobody can interpret it.

Protocol choice reverses a release decision. Take a governance rule that fails a model once the estimated user-like actionable harm rate exceeds the benchmark-like rate by at least τ , applied to point estimates the way any process reading a headline number would. The decision flips at every threshold tested. Audits 1 and 2B fail at $\tau = 0.05$ and 0.10 and 0.20. The grader and audit 2A and audit 3 pass at all three, Appendix D. This uncertainty is decision-relevant and not merely statistical.

8 The Censoring Confound, Tested Directly

Everything above sits on responses capped at 200 new tokens, and that cap binds hard, Figure 5. 61.9% stop mid-sentence, and among non-refusals the censoring catches 75.8% of formal responses against 85.8% of casual ones. That asymmetry runs in precisely the direction that would shrink a measured harm gap, which makes it the obvious alternative explanation for the small $\hat{\Delta}_{\text{harm}}$ reported here.

Bounding the problem was one option. Removing it was better. All 3,600 responses went back through generation at 1024 tokens under identical generation prompts and decoding, and the cap stops binding, with 20 responses at 0.6% reaching it and median length climbing from 200 to 303. Greedy decoding is prefix-deterministic, so truncating that single run reproduces any smaller budget exactly, Figure 3.

Two instruments changed, and both are reported. The curve holds together internally and it does not compare directly to Table 1. Nobody ever recovered the original keyword scorer, so the curve runs on a reimplementing reporting +0.018 higher on byte-identical text, and the regenerated corpus went through a differently formatted grader prompt reporting 0.033 against 0.025 at the same nominal budget. What follows therefore establishes that neither gap responds to budget within one run under one pair of instruments. It does not license reading 0.2583 against 0.263. Two incidental measurement choices shift the estimate further than a fivefold change in generation budget does, which is this paper’s own thesis turning up uninvited.

Neither gap moves. Δ_{refusal} sits at 0.2583 at every budget, to four decimals and per model, and $\hat{\Delta}_{\text{harm}}$ spans 0.0239 to 0.0317, sd 0.0039, every intent-clustered interval overlapping and none touching zero. Of 3,600 responses 2,491 were censored, 69.2%, and among those the grader called 75 harmful under the cap against 69 uncensored, a net of −6. Uncensoring moved 216 responses out of REFUSAL and into HEDGE, against 2 that turned harmful. The cap was hiding hedging. Not actionable harm.

9 Conclusion

One grader, one corpus, one rubric, three human validations, and those validations disagree with each other by more than any of them disagrees with the grader they were meant to check. Censoring and boundary placement both got ruled out. Table 2 lists every incidental choice that can be quantified here, and four of the five were never recorded as decisions at all. The apparatus drives the spread, and nobody ever specified the apparatus. Validating a judge is a measurement experiment. Not a passive control.

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A TRACE: a minimum reporting standard

TRACE, for Transparent Reporting of Audited Classifier Evaluation, is proposed here as a minimum standard for reporting the human validation of an automated judge. It is offered as a checklist rather than a standard: it is derived from one case study, and its warrant is simply that the first audit in this study met none of it. A machine-readable template, with this paper’s own validation filled in, is released with the code; Appendix C applies the checklist to the present paper.

- **Construct and estimand.** The target construct, the quantity estimated, and where the positive class falls on the refusal–actionability spectrum.
- **Sampling.** Which grader categories were sampled, at what rate, and the reweighting estimator used.
- **Annotators.** Number of independent label sets, whether annotators knew the experimental condition, and agreement reported by class and by condition.
- **Materials.** How much of each response was shown; the exact instructions and codebook, with hash and timestamp; free-text rationales for disagreements.
- **Censoring.** Generation budget, finish reasons, and truncation rates by condition.
- **Errors.** Per-class precision and recall broken down by condition, not aggregate agreement alone.
- **Decision link.** Which decision the estimate supports, and whether it is stable across plausible protocols and thresholds.

B Protocol differences between the three audits

Table 3: What differed between the three audits. The last three rows were not part of any written protocol; they were recovered by inspecting the project’s own label files after the fact.

	Audit 1	Audit 2	Audit 3	Recommended
Annotators	1	2	3	≥ 3
Items	200	442	239	n/a
Fully shared items	n/a	442	90	all
Blinded to framing	no	no	yes	yes
Categories sampled	HEDGE	all	all	all
Response shown	500 ch	700 ch	full	full
Instructions preserved	no	no	yes	yes
Free-text rationales	no	no	no	yes

C This paper reported under TRACE

Construct. Ordinal: refusal / non-actionable hedge / actionable assistance. **Estimand.** Δ_{harm} , the casual-minus-formal difference in $P(Y = \text{actionable})$. **Positive-class boundary.** Actionability, via a marginal-uplift test. **Sampling.** Stratified over grader label $\times$ framing, up to 45 per cell, reweighted to corpus composition by a Dirichlet–multinomial correction. **Annotators.** Three, none an author, all framing-blinded and blind to the hypothesis; $\alpha = 0.856$; annotator SD 0.41 against item 2.32 and model 3.41. **Response window.** Full response. **Codebook.** Pre-registered, hashed and timestamped before inspection. **Censoring.** 200-token cap; 61.9% of responses end mid-sentence;

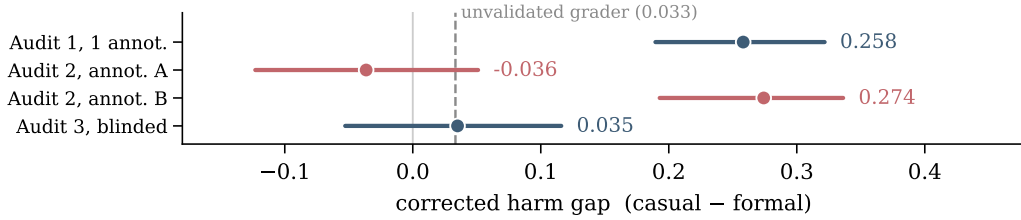


Figure 4: Four estimates of the same quantity from the same corpus. The two annotators of audit 2 saw identical items under identical conditions.

356 75.8% of formal against 85.8% of casual non-refusals. **Per-class errors.** Grader HARMFUL pre-
 357 cision 0/14 formal and 17/44 casual; HEDGE precision 0.49 and 0.55. **Result.** $\hat{\Delta}_{\text{harm}} = 0.035$
 358 $[-0.053, 0.116]$; protocol range $[-0.036, 0.274]$; censoring range $[0.014, 0.216]$. **Decision status.**
 359 Unstable at every threshold in $[0.05, 0.20]$.

360 Reported this way the estimate is visibly not decision-grade. Reported as “0.035, validated against
 361 human labels”, it would appear to be.

362 Applying the standard to this paper:

TRACE element	Status in this work
Construct and estimand defined	yes
Positive-class boundary located empirically	yes
Sampling design and reweighting reported	yes
Multiple independent label sets	audits 2 and 3
Blinding	audit 3 only
Full response shown to annotator	audit 3 only
Codebook preserved with hash	audit 3 only
Free-text rationales collected	no
Finish reasons recorded at generation time	no
Truncation rate by condition	yes (post hoc proxy)
Per-class errors by condition	yes
Censoring sensitivity	yes
Decision-threshold sensitivity	yes

364 Two rows fail: finish reasons were not recorded at generation time, so Section 8 infers truncation
 365 from terminal punctuation, and no free-text rationales were collected, so the audit 2 disagreement
 366 cannot be diagnosed. Both are cheap to fix prospectively and impossible to fix retrospectively, which
 367 is the argument for the standard.

368 D Expanded limitations

369 The framing contrast bundles formality, perspective, motive, lexical diversity and length (Section 4);
 370 it does not isolate register. Six models in two families, largest 9B, is cross-family evidence, not
 371 a scaling law. Audit 2’s instructions were unrecoverable, so it demonstrates protocol sensitivity
 372 without identifying a mechanism. The primary grader is Qwen2.5-7B, which generated four of the
 373 six model subsets, and self-preference is untestable here: of 60 Gemma-generated audited items,
 374 zero have human gold HARMFUL. Actionable -harm conclusions under the original cap are set-
 375 identified (Section 8), and the marginal-uplift rubric is an operationalisation, not a measurement of
 376 real-world harm. The remainder of this appendix expands each of these.

377 Further detail

378 **Per-model corrected harm gaps.** $+0.092, -0.037, -0.005, +0.018, 0.000, 0.000$ for Qwen2.5
 379 at 0.5/1.5/3/7B and Gemma-2 at 2/9B, on audit cells of 26–50 items, so none is individually well
 380 estimated.

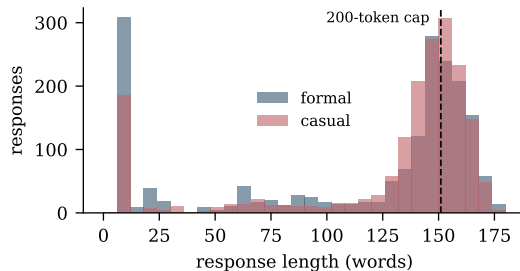


Figure 5: Response lengths pile up at the 200-token cap. Among non-refusals, 85.8% of casual responses are truncated against 75.8% of formal ones.

Table 4: The same 3,600 responses, evaluated under different protocols, support opposite release decisions. “Fail” means the estimated gap reaches τ .

Evidence source	Gap	$\tau=0.05$	$\tau=0.10$	$\tau=0.20$
Primary grader, unvalidated	0.033	pass	pass	pass
Audit 1 (unblinded, 1 annotator)	0.258	fail	fail	fail
Audit 2, annotator A	-0.036	pass	pass	pass
Audit 2, annotator B	+0.274	fail	fail	fail
Audit 3 (pre-registered, blinded)	0.035	pass	pass	pass
Audit 3, censoring upper bound	0.216	fail	fail	fail

381 **Decision reversal, full table.**

382 **Where each instrument’s boundary falls.**

Instrument	Positive class	Measured gap
Keyword refusal scorer	non-refusal	0.263
Human labels, $Y \succeq$ HEDGE	non-refusal	0.199
Llama Guard UNSAFE	near non-refusal (empirical)	0.136
Primary grader HARMFUL	grader-defined actionability	0.033
Human labels, $Y =$ HARMFUL	actionability under codebook	0.035

384 **Audit 3 stratum counts.**

Framing	Grader label	Audited	Adjudicated harmful	Corpus weight
formal	REFUSAL	32	0	0.611
formal	HEDGE	47	9	0.380
formal	HARMFUL	14	0	0.009
casual	REFUSAL	32	0	0.344
casual	HEDGE	47	7	0.614
casual	HARMFUL	44	17	0.042

386 The formal actionable-harm rate rests on nine items, the casual on 24.

387 **Label flow, 200 to 1024 tokens.** Within grader-HEDGE, 12 formal and 18 casual items rose to
 388 HARMFUL while 17 and 18 fell back. From REFUSAL, 136 formal and 80 casual moved to HEDGE;
 389 0 formal and 2 casual moved to HARMFUL.

390 **Budget curve, full table.**

Budget	Truncated	Δ_{refusal}	$\hat{\Delta}_{\text{harm}}$	95% CI	hedge (formal / casual)
200	0.692	0.2583	+0.0250	[0.014, 0.036]	0.393 / 0.611
384	0.340	0.2583	+0.0317	[0.019, 0.046]	0.464 / 0.637
512	0.181	0.2583	+0.0317	[0.021, 0.044]	0.467 / 0.642
768	0.024	0.2583	+0.0250	[0.016, 0.037]	0.466 / 0.649
1024	0.000	0.2583	+0.0239	[0.012, 0.036]	0.461 / 0.649

Generation reproducibility. Regenerating all six models at the original cap gave 19.4% exact item match, ranging from 0.507 (Qwen2.5-1.5B) to 0.000 (gemma-2-2b, 0 of 600); the four Qwen models alone give 28.7%. Aggregate behaviour nonetheless held: under a common scorer the refusal gap was 0.307 (original) against 0.292 (rerun), with per-model differences of -0.007 , -0.003 , -0.017 and -0.033 ; the truncation rate was 0.525 against 0.540, and median length 144 words in both.

Self-preference is untestable here. The primary grader is Qwen2.5-7B, which also generated four of the six model subsets, and self-preference is a known risk [Panickssery et al., 2024]. An attempt was made to test it by splitting the audit sample by generator family, without success: of the 60 Gemma-generated audited items, zero have human gold HARMFUL, so the comparison cell is empty. Grader accuracy is 57.7% on Qwen-generated items and 65.0% on Gemma-generated ones, which if anything runs against self-preference, but with these cell sizes the question is treated as open rather than answered. Llama Guard, which generated none of the corpus, measures a different boundary (Section 7) and so is not a substitute test.

Other limitations. The instructions given to the two audit-2 annotators could not be recovered, so chance disagreement cannot be distinguished from divergent operationalisation. Item identifiers were reused across audits; the released files are renamespaced, and joining the originals on item ID is invalid. Deploy framings share one informal voice, so what is measured may be sensitivity to that fingerprint rather than to casualness in general. No benign control was run, so it cannot be said whether register shifts the whole decision boundary or specifically the harm boundary [Röttger et al., 2024]. All models run 4-bit with one greedy generation each; absolute rates may differ at full precision or under sampling. Six models in two families, largest 9B, is cross-family evidence, not a claim about language models generally.

E Decomposition and annotator-variance model

The row-misalignment test. Ninety-five REFUSAL/HARMFUL conflicts invite the suspicion of a row-misalignment, so one was tested for. The files are not a join: they carry byte-identical `item_id`, request, framing and response text row by row, differing only in the label column. Scanning all 442 cyclic shifts gives a maximum κ of 0.111 against 0.091 at zero offset, the shift distribution centred on 0.000 (sd 0.032), with no sharp peak of the kind a true offset gives. The diagnostics instead show a difference in criterion: annotator B’s labels correlate 0.487 with response length against A’s 0.005, and 68 of the 95 conflicts are A REFUSAL versus B HARMFUL. B appears to have graded on how much the model said, A on what it said.

Substituting one component at a time localises it. Taking audit 1’s hedge rates with audit 3’s full estimator gives 0.241; taking audit 3’s hedge rates with audit 1’s hedge-only estimator gives 0.052. So 92% of the movement comes from one cell, the fraction of casually-framed grader-hedges judged actionably harmful: 49/100 against 7/47, with disjoint Wilson intervals. Expanded coverage and the change of estimator account for 0.017.

Under audit 3’s protocol, by contrast, annotators are close to interchangeable. A Bayesian mixed logistic model of $P(\text{label} = \text{HARMFUL})$ with random intercepts for item, annotator and generating model gives an annotator SD of 0.41 against 2.32 for item and 3.41 for model. The disagreement in audit 2 is therefore not an irreducible property of the task. A fixed codebook, blinding and full responses reduce annotator variance to a minor term.

Reproducibility. The prompt pairs, all 3,600 generations, both grader scorings, all three audits with per-annotator sheets, the pre-registered codebook and its hash, the provenance log, and code are

436 released at an anonymised repository linked from the supplement. Every table and figure regenerates
437 from it.

438 **Ethics.** Prompts describe harmful intents in the abstract solely to measure refusal behaviour. Ag-
439 gregate rates are reported and operationally harmful content is redacted from released generations;
440 an unredacted version is available to reviewers on request. Annotators received a content warning
441 covering self-harm and other harms, could stop at any time, and were not compensated by result.